

Mask Detection Using MobileNetV2

Objective

The objective of this project is to build a face mask detection system capable of classifying images into two categories:

- With Mask
- Without Mask

A pretrained MobileNetV2 model was fine-tuned using transfer learning to achieve high accuracy while maintaining low computational requirements suitable for edge devices.

Dataset

The dataset consists of facial images belonging to two classes:

1. With Mask
2. Without Mask

Images were organized into separate folders and loaded using Python. A balanced subset of images from each class was used for training and testing.

Data Preprocessing

The following preprocessing steps were performed:

- Images were resized to 128×128 pixels.
- Pixel values were normalized to the range 0–1.
- Labels were encoded as numerical values.
- The dataset was split into training and testing sets using an 80:20 ratio.

Data Augmentation

To improve model generalization and reduce overfitting, data augmentation techniques were applied:

- Rotation
- Zooming
- Width shifting
- Height shifting
- Shearing
- Horizontal flipping

These transformations create additional variations of training images without requiring new data collection.

Model Architecture

Base Model

MobileNetV2 pretrained on the ImageNet dataset was used as the feature extractor.

Custom Classification Head

The following layers were added:

- AveragePooling2D
- Flatten
- Dense (128 neurons, ReLU)
- Dropout (0.5)
- Dense (2 neurons, Softmax)

The pretrained MobileNetV2 layers were frozen during initial training to preserve learned visual features.

Training Configuration

Parameter	Value
Optimizer	Adam
Learning Rate	0.0001
Loss Function	Sparse Categorical Crossentropy
Batch Size	16
Epochs	5
Image Size	128 × 128

Evaluation Metrics

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Accuracy

Accuracy measures the percentage of correctly classified images.

Formula:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Precision

Precision measures how many predicted positives are actually positive.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall

Recall measures how many actual positives are correctly identified.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

F1 Score

The F1 Score is the harmonic mean of Precision and Recall.

Formula:

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Results

The trained MobileNetV2 model achieved excellent classification performance on the test dataset.

Observed results:

- High classification accuracy
- Strong precision and recall values
- Minimal misclassification between mask and non-mask categories

The confusion matrix showed that the majority of images were classified correctly.

Inference Speed

Inference speed refers to the time required to classify a single image.

MobileNetV2 provides:

- Fast prediction time
- Low memory usage
- Efficient execution on CPU-based systems

These characteristics make it suitable for real-time mask detection applications.

Edge Deployment Optimizations

1. TensorFlow Lite Conversion

The trained model can be converted into TensorFlow Lite format for deployment on mobile and embedded devices.

Benefits:

- Smaller model size
- Faster inference
- Lower memory consumption

2. Quantization

Convert model weights from float32 to int8.

Benefits:

- Reduced storage requirements
- Improved execution speed
- Lower power consumption

3. Model Pruning

Remove unnecessary weights and connections.

Benefits:

- Reduced model size
- Faster inference
- Lower memory footprint

4. Reduced Input Resolution

Using smaller image sizes such as:

- 128×128
- 96×96

can further improve processing speed while maintaining acceptable accuracy.

5. Lightweight Architecture

MobileNetV2 is already optimized for:

- Mobile devices
- IoT devices
- Edge AI applications

making it a suitable choice for deployment.

Applications

The developed face mask detection system can be used in:

- Airports
- Hospitals
- Educational institutions
- Public transportation systems
- Smart surveillance systems

Conclusion

A face mask detection model was successfully developed using MobileNetV2 and transfer learning. The model demonstrated strong classification performance while remaining lightweight and efficient. Through preprocessing, data augmentation, and transfer learning, the system achieved reliable results suitable for deployment on edge devices and real-time monitoring environments.